Area of Regular Polygons

Lesson 5-4

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A stop sign is a regular octagon — all 8 sides are the same length and all 8 angles are equal

Regular Polygon

A shape where all sides and all angles are equal.

Draw lines from the center of a hexagon to each corner → you get 6 equal triangles you can find the area of

Decompose

To break a shape into smaller, simpler shapes.

Each triangle inside a decomposed hexagon has a base = one side of the hexagon and height = distance from center to side

Triangle

A shape with three sides.

6 triangles, each with area 15 sq ft, combine to form a hexagon with total area = 6 × 15 = 90 sq ft

Composite

Made by putting two or more simple shapes together.

A house shape = rectangle (walls) + triangle (roof); find each area, then add them together

Composite figure

A shape made by putting two or more simple shapes together.

For a regular hexagon: Total Area = 6 × (½ × b × h), where b = side length and h = height of each triangle

Formula

A math rule written with symbols.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a regular polygon by decomposing it into triangles.

1️⃣ Notice

Your architecture firm is designing a hexagonal skylight for the lobby of a new museum.

2️⃣ Key idea

I can find the area of a regular polygon by decomposing it into triangles.

3️⃣ Words I'll use

Regular Polygon

Decompose **Triangle**

Composite

Composite figure

Formula

🤔 Think About It

- What shape is the skylight?
- How many sides does a hexagon have?
- Can you see smaller shapes inside the hexagon?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The hexagonal skylight has sides of 6 feet, and each triangle from the center has a height of 5.2 feet. Why is it helpful to break a hexagon into triangles to find its area?

Level 1 support

Start here: We decompose the hexagon into triangles because we already know how to find a triangle's area.

💡 **Need a hint?**

Sentence starters (optional)

We decompose the hexagon into ____ because ____.

Descomponemos el hexágono en ____ porque ____.

Once it is split, I can find each area using ____.

Una vez dividido, puedo hallar cada área usando ____.

Word bank: regular polygon | decompose | triangle | composite | formula

Listen for: *Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.*

Level 2

Predict whether this decompose-into-triangles strategy works for ANY regular polygon. Justify using the relationship between sides and triangles.

- This strategy ____ work for any regular polygon because ____.
- A regular polygon with n sides splits into ____ triangles, so ____.

EXPLORE

You matched each regular polygon to its number of center triangles. What pattern connects the number of SIDES to the number of TRIANGLES, and why?

Level 1 support

Start here: A regular polygon splits into the same number of triangles as it has sides.

Need a hint?

Sentence starters (optional)

I split each polygon into ___ triangles.

Dividí cada polígono en ___ triángulos.

The number of triangles equals the number of ___ because ___.

El número de triángulos es igual al número de ___ porque ___.

Then I added the triangle areas to get ___.

Luego sumé las áreas de los triángulos para obtener ___.

Word bank: regular polygon | decompose | triangle | composite | formula

Listen for: *A strong answer states the number of triangles equals the number of sides, because each side becomes the base of one triangle drawn to the center.*

Level 2

A classmate says you should ADD the 6 triangle areas, but another says MULTIPLY one area by 6. Are both correct? Justify when each works for a regular polygon.

- Both ___ correct because the triangles are ___.
- Multiplying works here since ___, but adding would be needed if ___.

CONNECT

The octagonal gazebo has 8 center triangles, each with base 5 ft and height 6 ft, and flooring costs $3 per sq ft. Walk through finding the total cost.

Level 1 support

Start here: Find one triangle's area, multiply by 8, then multiply by the price.

💡 **Need a hint?**

Sentence starters (optional)

One triangle's area is 1/2 times 5 times 6, which equals ___ square feet.

El área de un triángulo es 1/2 por 5 por 6, que es ___ pies cuadrados.

The octagon has 8 triangles, so the total area is ___ square feet.

El octágono tiene 8 triángulos, entonces el área total es ___ pies cuadrados.

The cost is ___ times $3, which equals $___.

El costo es ___ por $3, que es $___.

Word bank: regular polygon · decompose · triangle · composite · formula

Listen for: *Students compute one triangle = 15 sq ft, total = 8 x 15 = 120 sq ft, and cost = 120 x $3 = $360, explaining each step.*

Level 2

If the city switched the gazebo from an octagon to a hexagon but kept the same triangle base and height, would the flooring cost more or less? Generalize why.

- A hexagon would cost ___ because it has ___ triangles.
- In general, more sides means ___, so ___.

Guided Examples

Example 1

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle?

Solution: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 6 \times 5.2 = 15.6$ square feet.

Answer: A. 15.6 sq ft

Example 2

Using the triangle from the previous question, what is the total area of the hexagonal skylight?

Solution: Total area = 6 triangles × 15.6 sq ft = 93.6 square feet.

Answer: A. 93.6 sq ft

Example 3

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area?

Solution: Each triangle: $\frac{1}{2} \times 4 \times 3.5 = 7$ sq cm. Total: $6 \times 7 = 42$ sq cm.

Answer: A. 42 sq cm

Write About the Math

The Writing Revolution

I can explain my steps using the words regular polygon, decompose, triangle, and composite.

1. Kernel Sentence

subject + verb

Model: Regular Polygon is a shape where all sides and all angles are equal.
Polígono regular es una figura donde todos los lados y ángulos son iguales.

Write a kernel sentence about regular polygon. Use a subject and a verb.
Escribe una oración base sobre polígono regular. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Regular Polygon matters in math
Polígono regular importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Regular Polygon matters in math because ____.** *Polígono regular importa en matemáticas porque ____.*
but *pero*	**Regular Polygon matters in math, but ____.** *Polígono regular importa en matemáticas, pero ____.*
so *entonces*	**Regular Polygon matters in math, so ____.** *Polígono regular importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about regular polygon.
Di un hecho verdadero sobre regular polygon.

Regular polygon ____.

Question
Pregunta

Ask a question about regular polygon.
Haz una pregunta sobre regular polygon.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about regular polygon.
Muestra entusiasmo sobre regular polygon.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with regular polygon.
Dile a un compañero qué hacer con regular polygon.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

We decompose the hexagon into ____ because ____.
Descomponemos el hexágono en ____ porque ____.

Once it is split, I can find each area using ____.
Una vez dividido, puedo hallar cada área usando ____.

I split each polygon into ____ triangles.
Dividí cada polígono en ____ triángulos.

Try It

Solve on your own. Check the answer key when you are done.

1. A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area?

A. 42 sq cm

B. 21 sq cm

C. 84 sq cm

D. 14 sq cm

Show your work:

2. How many triangles can you make from the center of a regular pentagon?

A. 5

B. 3

C. 6

D. 10

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A regular hexagon and a regular pentagon both have triangles with the same base (6 ft) and the same height (5 ft). Which polygon has the greater total area? Explain why the number of sides matters.

Sentence starter: One triangle's area is ½ × ____ × ____ = ____. The hexagon has ____ triangles so its area is ____. The pentagon has ____ triangles so its area is ____. The ____ has more area because ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A regular pentagon is decomposed into 5 triangles from its center. Each triangle has a base of 7 inches and a height of 4.8 inches. What is the total area of the pentagon?

A. 84 sq in

B. 16.8 sq in

C. 168 sq in

D. 84 in

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. 42 sq cm *— Each triangle: ½ × 4 × 3.5 = 7 sq cm. Total: 6 × 7 = 42 sq cm.*
2. **Try It 2:** A. 5 *— A regular pentagon has 5 sides, so drawing lines from the center to each vertex creates 5 equal triangles.*
3. **Exit Ticket:** A. 84 sq in *— One triangle: A = 1/2 x 7 x 4.8 = 16.8 sq in. Total = 5 x 16.8 = 84 square inches.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Regular Polygon is a shape where all sides and all angles are equal.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).